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Paleomagnetism of Tertiary rocks on the Korean Peninsula: tectonic implications for the opening of the East Sea (Sea of Japan)

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Abstract

Two different distributions of declinations with dual polarities are found in the Tertiary rocks of the Yeonil Basin and Yangnam Basin along the Yangsan Fault in the southeastern part of the Korean Peninsula. These are northward declinations in the Middle Miocene Yeonil Group and strong eastward declinations in the Early Miocene Yangbuk Group and in the Eocene Volcanic Rocks. This is interpreted by invoking (1) the Yangsan Fault as a dextral wrench fault system by spreading in the Uleung Basin of the East Sea (Sea of Japan), and (2) a clockwise vertical rotation caused by the movement of the fault at around 17.3 Ma just prior to the sedimentation of the Yeonil Group. The timing of the deflection found in Korea is distinguished from that in southeastern and northwestern Japan at around 15 Ma. Two stages of spreading are thus constrained in the East Sea area: parallel at 23–15 Ma and pivotal at around 15 Ma. Crustal thinning also occurred. A reliable pole position (84.9ºN, 292.6ºE, $d_p/d_m = 3.3º/4.8º$) for the Miocene (about 15 Ma) in Korea is obtained from the Yeonil Group. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Tertiary; East Sea (Sea of Japan); tectonics; Korea

1. Introduction

The origin of the East Sea remains poorly understood despite the availability of extensive geologic and geophysical data along its coastal margin of Korea and Japan. From the paleomagnetic work for pre-Tertiary rocks, Kawai et al. (1971) noticed a remarkable difference in declination observed between southwestern (SW) and northeastern (NE) Japan. Such deflections extending into the Early Miocene constrain opening models of the East Sea (for example, Otofuji and Matsuda, 1983; Hayashida et al., 1991; Jolivet et al., 1991).

There are two typical Cenozoic basins, the northern Yeonil Basin and the southern Yangnam Basin, in the southeastern part of the Korean Peninsula (Pohang area). Considerable clockwise deflections of declination for Early Miocene volcanic rocks and the Eocene volcanic rocks have been reported from paleomagnetic data of the Yangnam Basin (Kim et al., 1986; Kim and Kang, 1989; Kikawa et al., 1994). It has been suggested that the deflections of the Yangnam Basin are caused by the clockwise rotation of the micro-blocks about a vertical axis by the dextral

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movement of the Yangsan Fault, which is thought to be related to the opening of the East Sea.

Much more paleomagnetic data (for example, for Cenozoic rocks in the Yeonil Basin) is needed to clarify rotational time, tectonic process and mechanism of the Pohang area. In this study, paleomagnetic results are presented for the Tertiary volcanic and sedimentary rocks from both the Yeonil and Yangnam Basins in the southern part of the Korean Peninsula in order to obtain more informations about the time of the clockwise rotation event of the area. We also obtained new paleomagnetic data for the Eocene Volcanic Rocks in the Yeonil Basin offering more concrete tectonic scheme of the area. The analysis of the results combined with previous data suggests a probable model of opening of the East Sea, which is interpreted as correlating with northward migration of the Philippine Sea Plate.

2. Geologic setting

In the Korean Peninsula, several Tertiary basins exist along the coast of the East Sea (Neogene and Paleogene) and the west coast region (Paleogene) of the Peninsula (Fig. 1). These are mainly composed of sedimentary (marine and nonmarine) and volcanic rocks. The typical basins intervened between the NNE–SSW-directional Yangsan Fault and the NW–SE-directional Ulsan Fault, are well exposed around the Pohang area (Fig. 1). The basins can be mainly divided into the southern (Yangnam Basin) and northern (Yeonil Basin) parts of the Pohang area by their stratigraphy and lithology (Yoon et al., 1991).

In the Yangnam Basin, the volcanic and nonmarine sedimentary rocks show relatively much more folded and faulted than the northern part. The isotopic age dating of the volcanic rocks in this area shows two violent volcanic events between 57 and 43 Ma (Eocene Volcanic Rocks) and between 23 and 17 Ma (Early Miocene Yangbuk Group) as shown in Table 1 (Lee and Pouclet, 1988; Jin et al., 1988; Lee et al., 1992). The Yangbuk Group is divided into the Janggi and Beomgokri subgroups which are composed of nonmarine sedimentary rocks, volcanic rocks and volcanic clasts (Tateiwa, 1924; Yoon et al., 1991). These subgroups cannot be correlated stratigraphically between sites due to insufficient data and lateral phase changes (Lee et al., 1992).

Marine formations of the Yeonil Group outcropping in the Yeonil Basin are deposited on the Eocene Volcanic Rocks and the Yangbuk Group. Byun (1995) provided a biostratigraphy based on dinoflagellate cysts from seven oil-exploration cores (300 to 700 m depth from top to basement rocks), and distinguished four zones: a barren zone (17.5–17.0 Ma), a *Distatodinium craterum* zone (17–16.3 Ma), a *Cribroperidinium giuseppei* zone (16.3–14 Ma), and a *Systematophora placacantha* zone (14–12 Ma). Yun et al. (1997) interpreted that the lower part of the Cheonbuk Conglomerate Formation is assigned to the barren zone, the upper part of the Cheonbuk Conglomerate Formation and the lower part of the Hakjeon Formation to the *Distatodinium craterum* zone, the upper part of the Hakjeon Formation to the *Cribroperidinium giuseppei* zone, and the Duho Formation to the *Systematophora placacantha* zone (Fig. 2). Formational times of the lower and the upper boundary of the Yoenil Group according to other recent biostratigraphic data are 17–12 Ma (nannofossils; Yi and Yun, 1995), 18.2–10.8 Ma (foraminifers; Kim, 1990), and 16.6–10.4 Ma (mollusca; Yoon, 1979) as illustrated in Fig. 2.

3. Paleomagnetic sampling and laboratory procedures

Seven to nine oriented block samples were collected at each site (Fig. 2 and Table 1). The natural remanent magnetization (NRM) of the specimens was measured using a two-axes cryogenic (SCT C-112) and a spinner (Schonstedt SSM-1) rock magnetometer. Detailed progressive thermal and alternating field demagnetization and rock magnetic studies were carried out. Alternating field (AF) demagnetization of specimens was performed in a laboratory-made apparatus with a three-axes tumbling device. The rock magnetic studies were thermomagnetic analyses in a saturating field of 0.85 T, acquisition of IRM (isothermal remanent magnetization), and thermal or AF demagnetization of the orthogonally induced IRM in a maximum field of 1.3 T. For more detailed descriptions, see Lee et al. (1996).

Two or more pilot specimens were stepwise de-

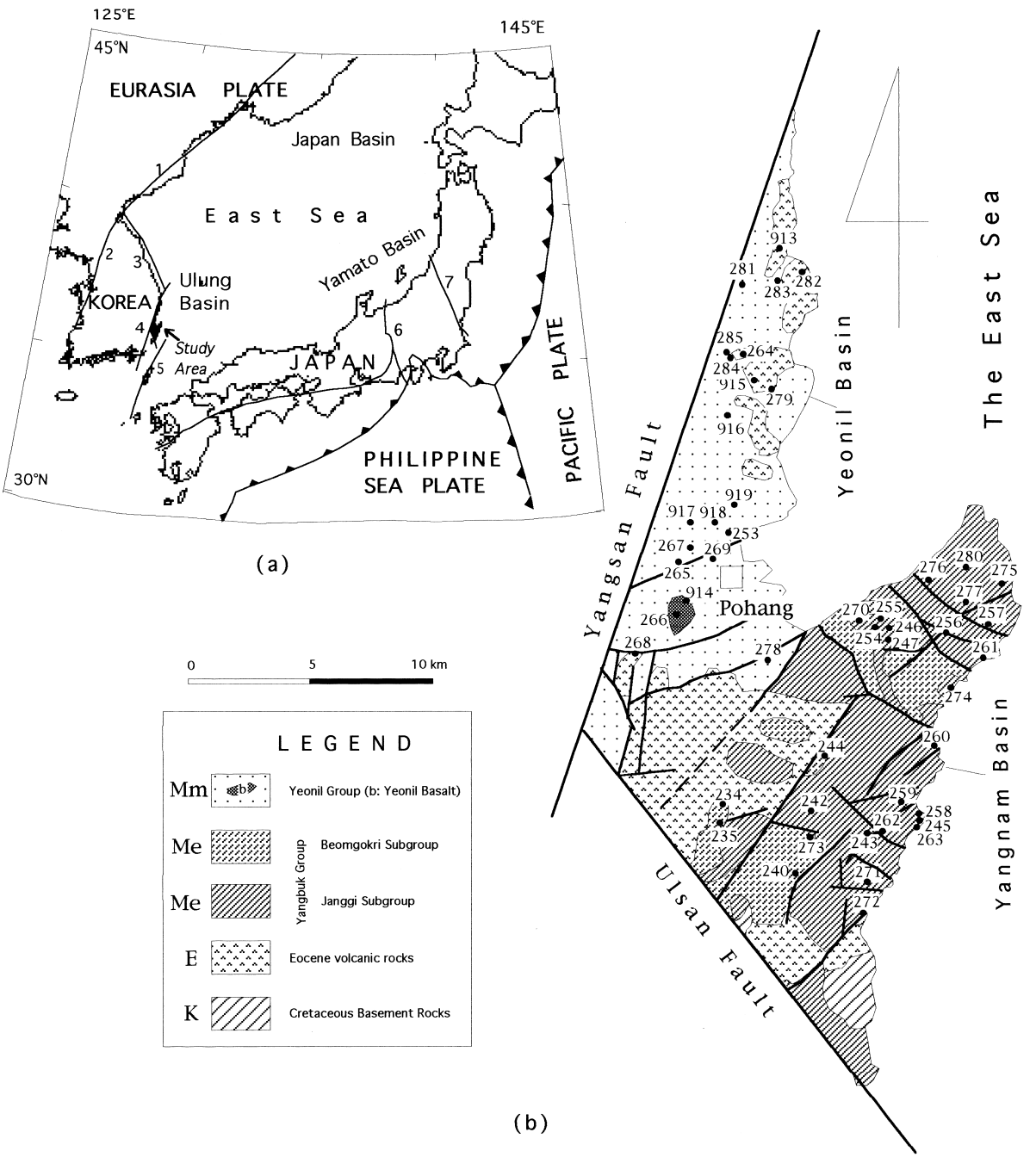


Fig. 1. (a) Tectonic map of the vicinity of the East Sea: *1* = Gilju–Myeongcheon Graben; *2* = Chugaryeong Fault; *3* = East Sea Fault; *4* = Yangsan Fault; *5* = Tsushima–Goto Fault; *6* = Itoigawa–Shizuoka Tectonic Line; *7* = Tanakura Fault. (b) Generalized geologic map of Pohang area (modified from Lee, 1987; Yoon et al., 1991; Yun, 1986) with sampling sites. *K* = Cretaceous; *E* = Eocene; *Me* = Early Miocene; *Mm* = Middle Miocene.

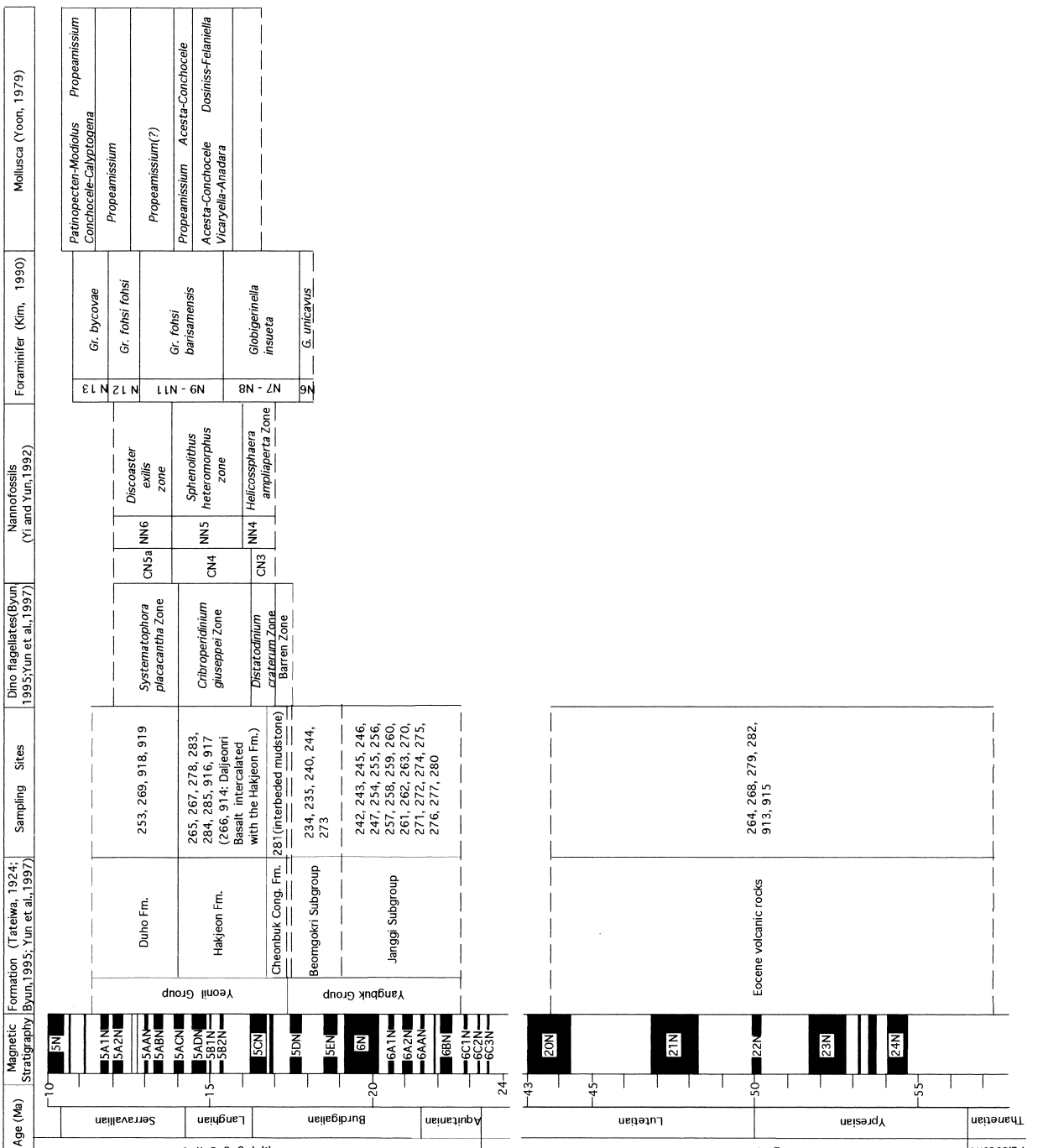
Table 1
Site mean directions from the Pohang area, Korea

	Site	n (N)	D_g	I_g	D_s	I_s	α_{95} (α'_{95})	k (k')	Lithology	Age-site (Ma)	Age-Fm. (Ma)
Duho Fm.	253	5	18.7	64.5	10.4	57.9	7.9	95.9	mudstone		
	269	4	−5.8	57.6	−19.2	43.2	14.9	38.9	mudstone		
	918	6	165.5	−18.2	176.3	−56.1	13.4	26.1	mudstone		
	919	5	−15.9	59.4	−2.5	44.6	14.1	30.2	mudstone		
	mean	(4)	−7.1	51.0	−4.9	50.9	12.3 (27.1)	56.4 (12.5)			
Hakjeon Fm.	278	5	−27.2	72.6	−5.8	69.3	25.6	9.9	mudstone		
	283	5	−13.5	41.5	−4.9	40.0	25.6	9.9	siltstone		
	284	8	−30.1	31.7	−20.0	47.8	18.6	9.8	mudstone		
	285	8	−9.0	42.4	10.4	51.8	9.5	35.1	mudstone		
	916	7	123.6	−41.8	178.4	−50.8	10.8	32.4	mudstone		
	917	7	175.9	−20.6	178.3	−54.5	12.3	25.1	mudstone		
	mean	(6)	−22.2	43.0	−4.8	52.5	9.2 (19.2)	53.7 (13.1)			
Yeonil Basalt	266	8	−15.4	51.3	−15.4	51.3	3.4	271.3	basalt	14.82 ± 0.40 [a]	
	914	6	−1.2	52.9	−1.2	52.9	5.0	180.0	basalt	15.16 ± 0.42 [a]	
	mean	(2)	3.8	54.8	−11.2	59.9	39.7 (54.9)	41.6 (22.8)			
Cheonbuk Fm.	281	6	−2.8	49.3	12.4	54.2	11.5	35.0	shale		
Beorngokri Fm.	234^	8	57.2	68.8	–	–	9.7	26.2	andesite		17.0 ± 0.7 [b]
	235^	5	51.9	61.4	–	–	7.6	68.2	andesite		22.4 ± 0.6 [b]
	240^	8	224.6	−40.8	–	–	3.7	180.8	andesitic tuff		15.93 ± 1.93 [c]
	244^	7	45.5	44.5	–	–	14.2	42.6	andesite		
	mean	(4)	48.4	54.0	–	–	15.8	34.8			

Table 1 (continued)

	Site	n (N)	D_g	I_g	D_s	I_s	α_{95} (α'_{95})	k (k')	Lithology	Age-site (Ma)	Age-Fm. (Ma)
Janggi Subgroup	242^	7	61.9	68.7	–	–	5.0	140.8	basalt	18.35 ± 0.62 [a]	18.74 ± 0.55 [a]
	243^	8	45.1	66.4	–	–	7.2	46.8	sandstone	19.6 ± 0.3 [c]	19.23 ± 0.63 [a]
	245^	5	61.4	66.5	–	–	13.3	22.2	basalt	17.99 ± 1.13 [a]	19.30 ± 0.50 [a]
	246	7	199.2	−20.0	214.5	−43.1	3.9	184.2	basalt	22.05 ± 0.67 [a]	19.92 ± 0.47 [a]
	247^	6	83.4	76.8	–	–	6.0	110.0	basalt	22.05 ± 0.67 [a]	20.36 ± 1.69 [a]
	257	4	14.6	52.3	30.4	56.7	10.0	86.2	andesite	22.7 ± 0.8 [e]	20.44 ± 0.47 [a]
	260	5	10.5	70.5	14.8	52.6	3.5	475.0	basaltic tuff		20.55 ± 0.67 [a]
	261^	5	199.6	−57.7	–	–	3.7	433.0	andesite		21.76 ± 2.27 [c]
	262	5	53.5	55.7	48.8	53.5	3.0	653.4	basalt	19.6 ± 0.3 [c]	19.3 ± 1.0 [d]
	263	5	50.4	26.6	47.7	46.2	3.2	571.6	rhyodacitic tuff		20.5 ± 0.8 [e]
	270*	8	42.8	21.4	34.3	11.2	6.1	84.1	basaltic tuff		
	274	5	33.9	7.5	40.9	29.9	2.9	699.0	basalt		
	275	4	61.8	61.7	44.2	81.1	6.3	212.3	andesitic tuff		
	276	2	183.1	−50.5	182.4	−40.5	36.7	48.5	andesite		
	277*	7	67.2	46.4	45.5	2.2	3.4	312.9	rhyolitic tuff		
	280	2	189.9	−33.5	199.0	−40.1	3.0	6927.0	rhyolite		
	mean	(14)	33.2	53.3	34.8	57.1	9.2 (12.7)	19.6 (10.8)			
Eocene Volcanic Rocks	264	7	30.4	56.1	56.4	47.0	5.0	148.1	rhyodacitic tuff	45.0 ± 1.1 [f]	43.73 ± 1.05 [a]
	279	3	201.6	54.2	87.5	70.1	6.0	418.5	rhyodacitic tuff	49.03 ± 1.09 [a]	44.03 ± 0.98 [a]
	913	7	282.9	−41.1	225.2	−55.2	16.1	14.9	welded tuff	53.2 ± 6.4 [f]	46.48 ± 2.47 [c]
	915	6	22.5	37.4	56.1	54.2	16.5	117.5	welded tuff	49.58 ± 1.13 [a]	57.32 ± 5.88 [c]
	mean	(4)	35.9	62.6	58.1	57.3	14.7 (51.7)	40.0 (4.1)			

Two mean directions marked * behind the site were excluded from the formation mean, and nine mean directions marked ^ behind the site were not available for obtaining bedding corrected values for no appearence of bedding attitude in the outcrop. Reference sources for K–Ar age data are as follows: [a] Lee et al. (1992), [b] Lee and Pouclet (1988), [c] Jin et al. (1988), [d] Tamanyu (1985), [e] Kim et al. (1986), [f] Shin and Jin (1996). n (N) represents number of specimens (sites). D_s (D_g), I_s(I_g), α_{95} (α'_{95}), and k (k') stand for declination, inclination, cone of confidence, and precision parameter after tilt correction (in situ), respectively. Age-site (Age-Fm.) denotes radiometric age reported from the site (formation).


Age (Ma)
Magnetic Stratigraphy
Formation (Tateiwa, 1924; Byun,1995; Yun et al.,1997)
Sampling Sites
Dino flagellates(Byun, 1995;Yun et al.,1997)
Nannofossils (Yi and Yun,1992)
Foraminifer (Kim, 1990)
Mollusca (Yoon, 1979)
Miocene
Paleocene
Eocene
Serravallian
Langhian
Burdigalian
Aquitanian
Thanetian
Ypresian
Lutetian
Yeonil Group
Yangbuk Group
Duho Fm.
Hakjeon Fm.
Cheonbuk Cong. Fm.
Beomgokri Subgroup
Janggi Subgroup
Eocene volcanic rocks
253, 269, 918, 919
265, 267, 278, 283, 284, 285, 916, 917 (266, 914: Daljeonri Basalt intercalated with the Hakjeon Fm.)
281(interbeded mudstone)
234, 235, 240, 244, 273
242, 243, 245, 246, 247, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 270, 271, 272, 274, 275, 276, 277, 280
264, 268, 279, 282, 913, 915
Systematophora placacantha Zone
Cribroperidinium giuseppei Zone
Distatodinium crateum Zone
Barren Zone
CN5a
CN4
CN3
NN6
NN5
NN4
Discoaster exilis zone
Sphenolithus heteromorphus zone
Helicosphaera ampliaperta Zone
N13
N12
N9 - N11
N7 - N8
N6
Gr. bycovae
Gr. fohsi fohsi
Gr. fohsi barisamensis
Globigerinella insueta
G. unicavus
Patinopecten-Modiolus Conchocele-Calyptogena Propeamissium
Propeamissium
Propeamissium(?)
Propeamissium Acesta-Conchocele
Acesta-Conchocele Vicaryella-Anadara Dosiniss-Felaniella
5N
5A1N
5A2N
5AAN
5ABN
5ACN
5ADN
5B1N
5B2N
5CN
5DN
5EN
6N
6A1N
6A2N
6AAN
6BN
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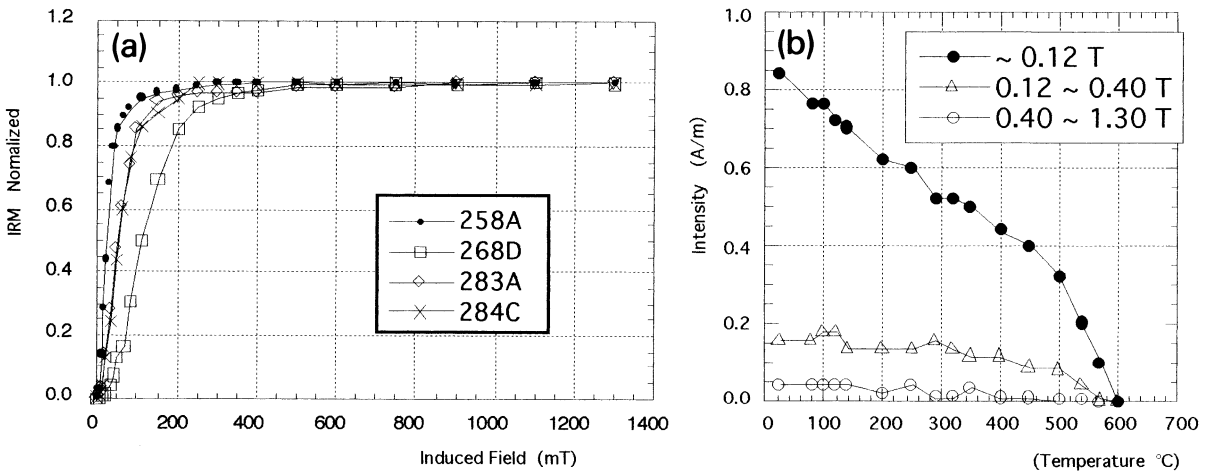


Fig. 3. Typical examples of (a) IRM (isothermal remanent magnetization) acquisition, and (b) progressive thermal demagnetization (sample 284C) curves.

magnetized first by using AF method for each site. Three pilot specimens of terrestrial sedimentary rocks (Site 243) were subjected to a progressive thermal demagnetization after an initial AF cleaning. Low-field susceptibility was measured using a Bartington MS2 susceptibility meter after each step of the PThD in order to detect chemical changes during the heating and cooling process. According to the results of the preliminary progressive demagnetization, all other specimens were subjected to progressive AF demagnetizations (4–5 steps from the suitable demagnetization level), while five specimens of Site 243 were subjected to thermal demagnetizations (11–13 steps from room temperature). All progressive demagnetization data were analysed using the principal component analysis method (Kirschvink, 1980).

4. Paleomagnetic result and analysis

Based on the determination of blocking temperature by thermal demagnetization and of IRM tests, titanomagnetite is identified as the main magnetic carrier, while hematite is distributed as accessaries in the rock samples. Acquisition of IRM (Fig. 3a) saturation at less than 500 mT indicates that titanomagnetite is a common magnetic mineral in the various types of basalt (sample 258A, the Yangbuk Group), tuff (sample 268D, the Yangbuk Group), and mudstone (samples 283A and 284C, the Yeonil Group). Fig. 3b is a typical example of demagnetization behavior of IRM in mudstone of the Yeonil Group (sample 284C).

Thermomagnetic curves (Fig. 4a) support the above IRM results. Irreversible heating and cooling cycles are generally seen. Some of them had a peak after 440°C heating (Fig. 4b), which may imply that iron sulfide minerals, such as pyrrhotite or pyrite, oxidized and converted to magnetite.

Two or more pilot specimens were stepwise demagnetized by using AF or the thermal method for each site. Paramagnetic minerals as well as pyrrhotite occasionally made strong VRM (viscous remanent magnetization), and screened the characteristic direction of titanomagnetite during thermal or AF demagnetization. Spurious magnetization acquired during AF demagnetization steps higher than 30–40 mT, was detected in some acidic volcanic and sedimentary rocks.

The typical behavior of the NRM in demagnetization is illustrated in Fig. 5. In Fig. 5a, a single

Fig. 2. Stratigraphy and sampling sites of the Pohang area, Korea. Magnetostratigraphy is from Harland et al. (1989). Note that the lowest ages of biostratigraphaphy obtained from the Yeonil Group represent a time range of 16.5 to 18.2 Ma.

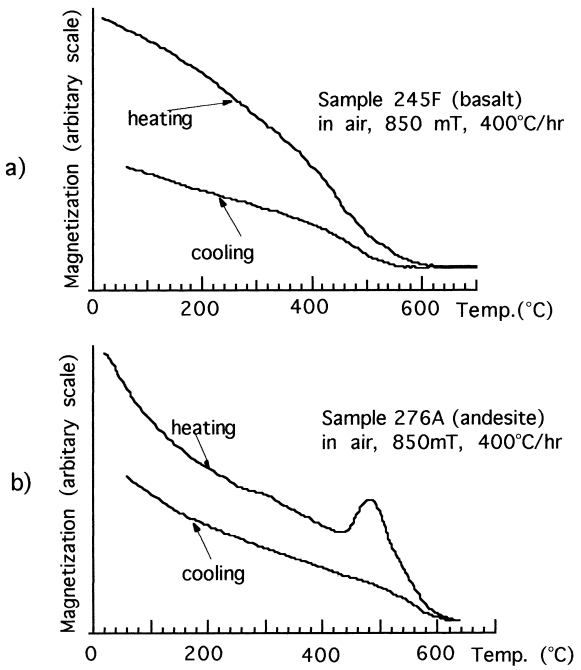


Fig. 4. Thermomagnetic curves of typical examples from the Pohang area. Both samples were magnetized with 0.85-T induction, and heated and cooled in air. (a) Single titano-magnetite phases are shown by the Curie temperature of about 560ºC. (b) An anomalous increment of magnetization at about 440ºC indicates that some chemical changes occur in heating.

component from specimen 253B (mudstone in the Yeonil Group) is seen after 5 mT. In another example of alternating field demagnetization (Fig. 5b, specimen 261A, andesite), a soft component of remanent magnetization is removed from a hard component after 20 mT. Hematite magnetization was found in some terrestrial sedimentary rock specimens from the Yangbuk Group (Fig. 5c, specimen 243D, sandstone). Its magnetization might be DRM (depositional remanent magnetization) derived from ambient basement of the widely distributed Cretaceous red beds.

Anomalous demagnetization behavior is shown in some sedimentary rocks of the Yeonil Group and in some tuffaceous rocks of the Yangbuk Group, because of their weak or unstable remanences (Fig. 5d, specimen 256A, tuff).

Paleomagnetic results were successfully obtained from 35 sites (11 sites of the Yeonil Group, 2 sites of the Yeonil basalt, 4 sites of the Beomgokri Subgroup, 14 sites of the Janggi Subgroup, and 4 sites of the Eocene Volcanic Rocks), and are shown in Fig. 6 and listed in Table 1. The site mean directions from the younger Yeonil Group (squares) are indistinguishable from the present mean axial geomagnetic field, while those from the older Yangbuk Group and the Eocene Volcanic Rocks (circles) are significantly deflected in declination.

We could not find any reliable mean directions from 12 sites due to the combination of unstable remanences, weak NRM intensities, absence of systematic demagnetization behavior, or hard VRM acquired during demagnetization as shown in Fig. 5d. Two mean directions of sites 270 and 277 (marked by an asterisk in Table 1) were discarded because of suspicions of remagnetization.

The stratigraphic correlations between the microblocks in this area bounded by faults are still controversial. Recently, however, several biozonation studies for the Yeonil Group have provided a reliable biostratigraphy in this area (Fig. 2). These data yield the mean times for the lower and the upper Yeonil Group of 17.3 $\pm$ 0.7 Ma (standard error at the 95% confidence level) and 11.3 $\pm$ 0.8 Ma, respectively. We used the mean value of biostratigraphic data (Fig. 2) for age correlation of the Yeonil Group, and K–Ar age data for various volcanic rock types of the Yangbuk Group and Eocene volcanic rocks (Tamanyu, 1985; Kim et al., 1986; Lee and Pouclet, 1988; Jin et al., 1988; Shin, 1992; Lee et al., 1992).

In this study, both normal and reversed polarities are found in the characteristic directions in the group (or subgroup) as shown in Table 1. The bedding attitude was found in 26 sites (Table 1), and the k values for site mean directions significantly increase after unfolding. Ratios k_2/k_1 (after/before bedding correction) for the upper Yeonil Group (4 sites in the Duho Fm.), lower Yeonil Group (7 sites in the Cheonbuk and Hakjeon Fms.), lower Yangbuk Group (9 sites in the Janggi Subgroup), and Eocene volcanic rocks (4 sites in the Yeonil and Yangnam Basins) are 4.51, 3.65, 2.03, and 9.76 respectively (Table 2), which are statistically significant at the 95% confidence level except the Janggi subgroup. Although the fold test for the Janggi subgroup was not significant at the 95% level, the site mean direction is interpreted as the primary one for the following reasons. (1) The increase of the ratio in the precision

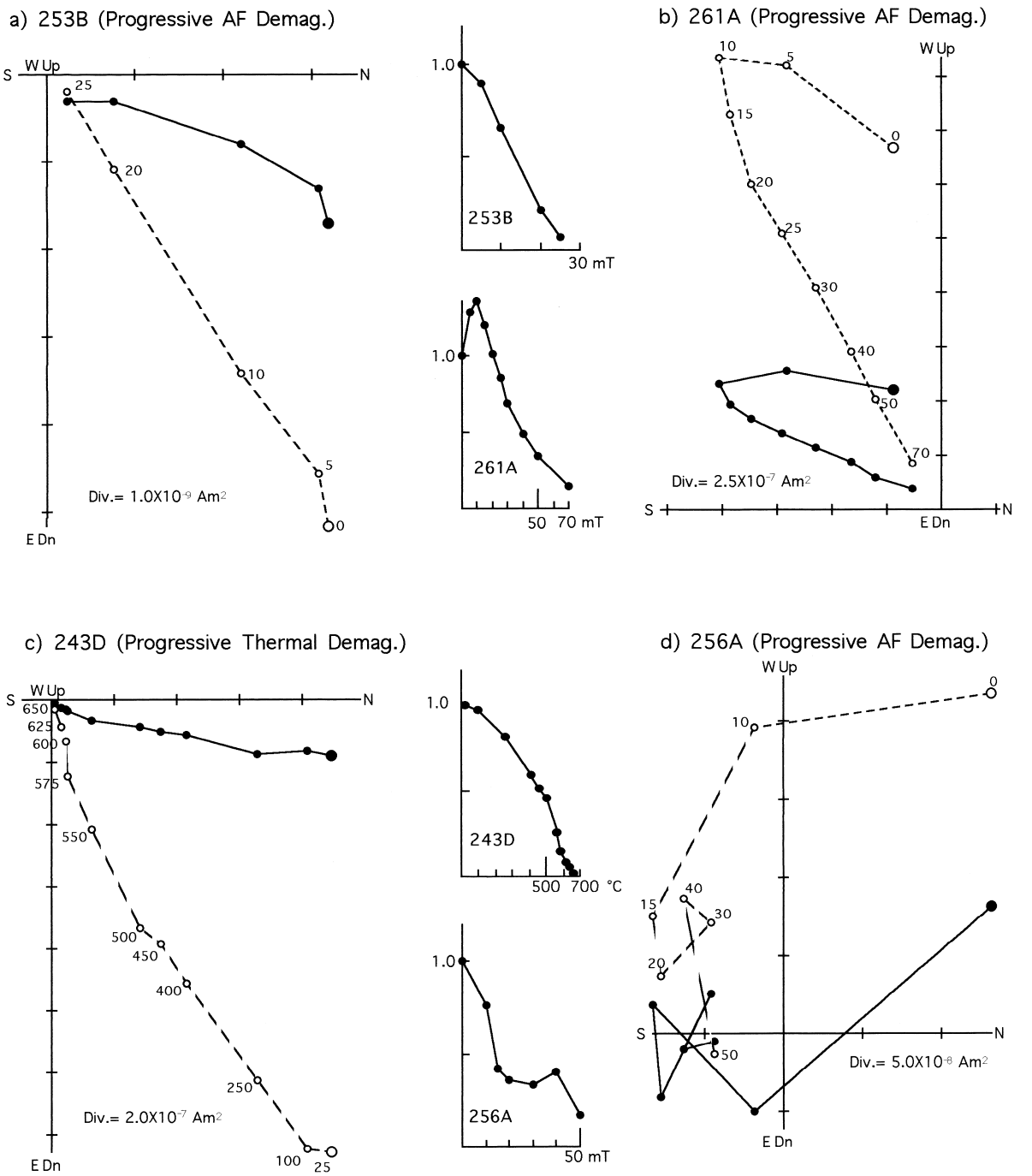


Fig. 5. Typical demagnetization vector and normalized demagnetization diagrams (in situ) for samples of (a) 253B (mudstone), (b) 261A (andesite), (c) 243D (sandstone), and (d) 256A (tuff).

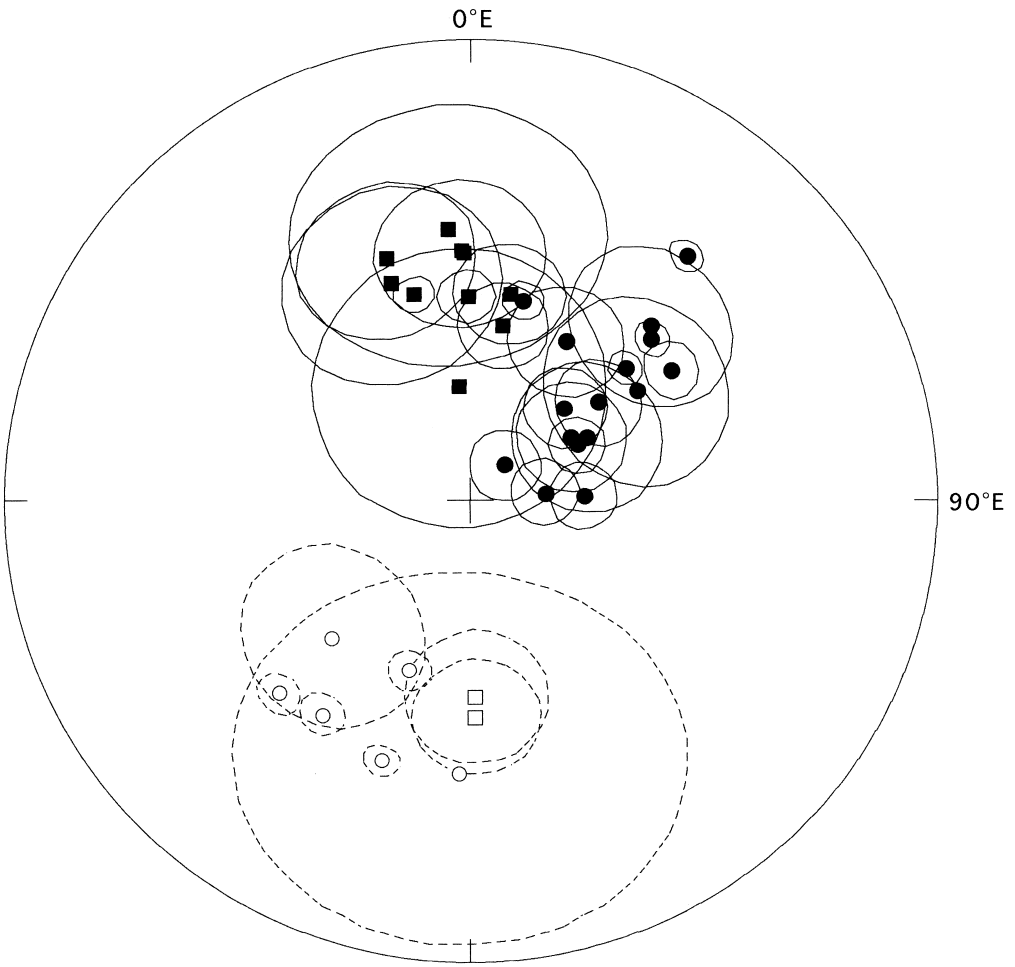


Fig. 6. Site mean directions with α_{95} circles of confidence for the Yeonil Group (squares) and the Yangbuk Group and Eocene Volcanic Rocks (circles) in the Pohang area (listed in Table 1). Open and solid symbols are projections on the upper and lower hemisphere, respectively.

Table 2
Formation mean directions and fold test results for well bedded site in this study

Group	N	D_g	I_g	α_{95g}	k_1	D_s	I_s	α_{95s}	k_2	k_2/k_1	Polarity
The Yeonil Group											
Upper Yeonil G.	4	−7.1	51.0	27.1	12.5	−4.9	50.9	12.3	56.4	4.51	N/R
Lower Yeonil G.	7	−19.4	44.2	16.3	14.7	−1.8	53.1	8.3	53.6	3.65	N/R
The Yangbuk Group											
Janggi subg.	9	28.4	43.8	17.0	10.2	30.1	50.2	11.6	20.7	2.03	N/R
The Eocene Volcanic Rocks											
(Yeonil and Yangnam Basins)	4	35.9	62.6	51.7	4.1	58.1	57.3	14.7	40.0	9.76	N/R

N, D_g (D_s), I_g (I_s), α_{95g} (α_{95s}), and k_1 (k_2) represent number of sites, declination, inclination, cone of confidence, and precision parameter in situ (after bedding correction), respectively.

parameter after/before unfolding ($k_2/k_1 = 2.03$). (2) Both normal and reversed polarities are found. (3) Neither in-situ nor tilt-corrected directions are identical to any post-Miocene direction, indicating that remagnetization is unlikely.

The present results with six previous studies from this area (Kim et al., 1986, 1993; Kim and Kang, 1989; K.H. Kim et al., 1994; Kim and Doh, 1994; Kikawa et al., 1994) are illustrated in Fig. 7. The deflection of declination indicates a significant event between the formations of the Yeonil and Yangbuk Groups at about 17.3 Ma, whilst there is no significant difference of inclination components between the groups (Fig. 7). In this study, the clockwise deflections of declination are newly obtained from the Eocene volcanic rocks in the Yeonil Basin implying that the deflections are found in both the Yangnam and Yeonil Basins. Such block rotation is not found in the western part of the Yangsan Fault or in southern part of the Ulsan Fault (Lee et al., 1997). Therefore a vertical-axis clockwise block rotation can be caused by dextral movement of the Yangsan Fault at 17.3 Ma before the formation of the Yeonil Group.

The range of declination/inclination observed from the Yangbuk Group and Eocene volcanic rocks is significantly larger than that from the Yeonil Group (Fig. 7, Table 3). These could partly be due to (1) the different amounts of vertical-axis rotation from site to site, (2) secular variation especially in volcanic outcrops, and (3) the unresolved bedding attitudes.

The directions of 35 sites in Yangbuk Group (Ync) and 14 sites in the Eocene volcanic rocks (Enc) cannot be corrected for the dip of the layers because of few appearances of bedding attitudes (for example, welded structure, intercalated sedimentary or volcanic ash layers) in the field. The tilt-corrected directions of 48 sites in the Yangbuk Group (Ytc) and 12 sites in the Eocene volcanic rocks (Etc) were compared with those of the non-corrected directions, indicating that the deflections are not simply due to the deviation of the absence of stratigraphic correction. The mean deviations of the declination (between 41.9ºE in Ync direction and 35.1ºE in Ytc direction) and the inclination (between 50.2º in Ync direction and 53.2º in Ytc direction) are dominant in the Yangbuk Group. The Ync direction (41.9ºE/50.2º, $k = 12.0$) is consistent with the Ytc (35.1ºE/53.2º, $k = 19.3$) after correction of the bedding attitude about N9.3ºE/6.1ºNW. For the Eocene volcanic rocks, there are deviations of the mean directions between Enc (57.3ºE/55.3º, $k = 15.8$) and Etc (38.0ºE/60.1º, $k = 16.7$). The Enc direction agrees with the Etc direction after bedding correction of N26.3ºE/13.0ºNW. This shows that a general bedding trend of a northwest dip can be expected in the Early Miocene Yangbuk Group and the Eocene volcanic rocks indicating no significant tectonic disturbance or deformation between the Eocene and Early Miocene. However, some site mean directions in the Yangbuk and the Eocene volcanic rocks are scattered after a bedding correction around the group mean direction. This may be caused by (1) the initial topographic gradient during extrusion of volcanic lava and tuff, (2) vertical axis block rotations involving deformation, (3) the fact that additionally more than two times of folding or deformation took place in the area, and (4) occasional remagnetization.

In addition, a pole position (84.9ºN/292.6ºE, $d_p/d_m = 3.3º/4.8º$) for the Miocene (about 15 Ma) in

Table 3
Group mean of tilt-corrected versus non-tilt-corrected directions in the Pohang area

	Yeonif Group			Yangbuk Group			Eocene Volcanic Rocks		
	Ltc	Lnc	L	Ytc	Ync	Y	Etc	Enc	E
Ns	22	1	23	48	35	83	12	14	26
mean (D/I)	1.7/50.4	−37/56.1	1.5/50.8	35.1/53.2	41.9/50.2	38.0/52.0	38.0/60.1	57.3/55.3	49.0/57.9
error (D/I)	4.5/2.7	–	4.4/2.7	7.3/3.3	6.5/6.8	5.1/3.4	17.3/7.4	17.0/4.2	12.2/4.2
α_{95}/k	3.6/74.2	–	3.5/75.6	4.8/19.3	7.3/12.0	4.1/15.3	10.9/16.7	10.3/15.8	7.4/15.6
k_2/k_1	2.9	–	–	1.8	–	–	1.9	–	–

N_s and error stand for number of sites and standard error for 95% confidence. Ltc (Ytc, Etc), Lnc (Ync, Enc), and L (Y, E) represent the tilt-corrected, non-tilt-corrected, and total directions of the Yeonil Group (Yangbuk Group, Eocene Volcanic Rocks).

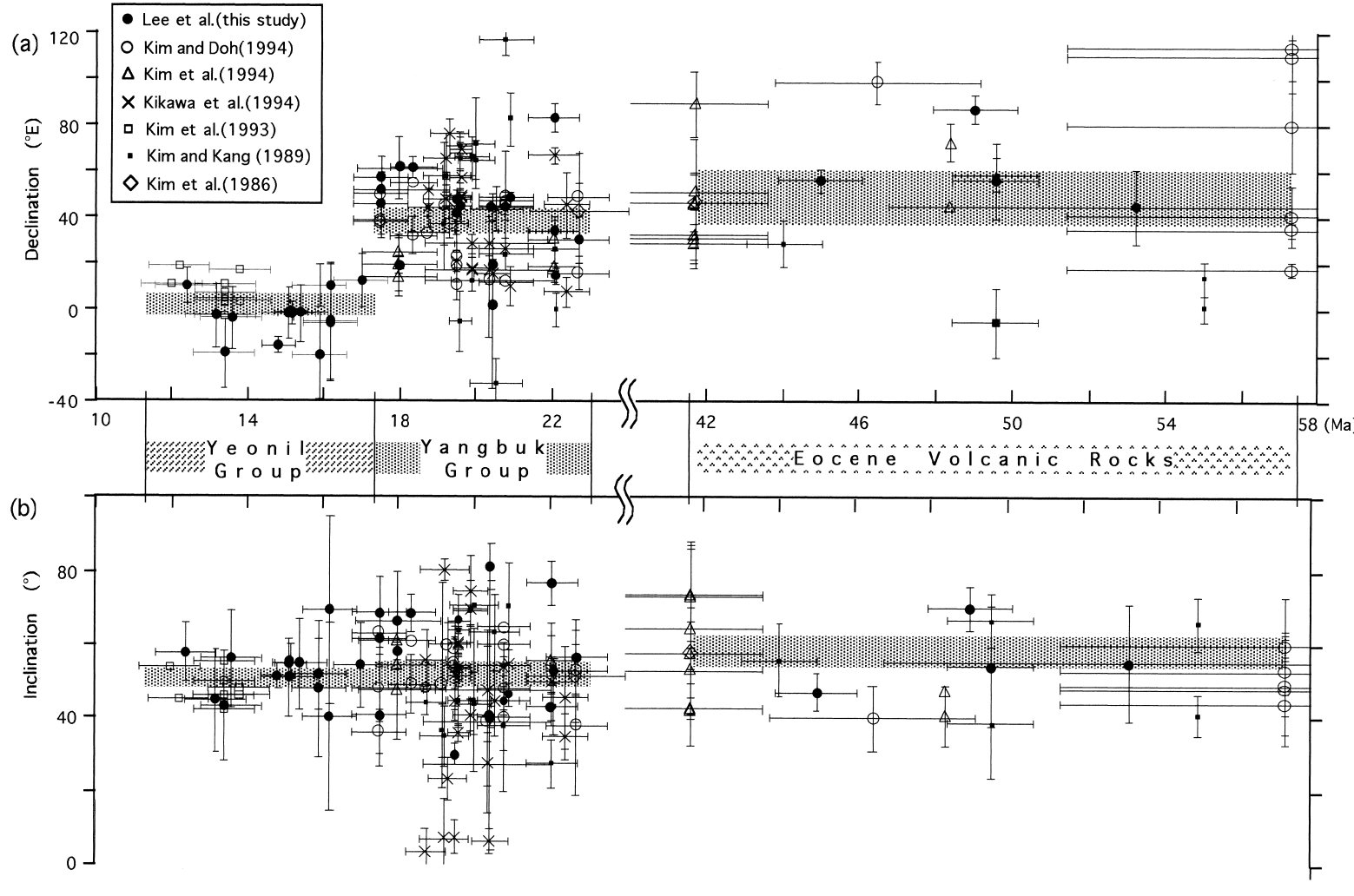


Fig. 7. The deflection diagrams for (a) declination and (b) inclination of the Tertiary rocks in the Pohang area. Each shaded zone represents the standard error range of 95% confidence for the group mean.

Korea is calculated from the paleomagnetic direction from 22 sites of the Yeonil Group.

5. Modes and timings of the East Sea opening

The Early Miocene clockwise deflection of the declination for SW Japan has been reported in many paleomagnetic studies (Otofuji et al., 1985, 1991; Hayashida et al., 1991; Ishikawa and Tagami, 1991). The fan-shape model of clockwise rotation is well known in the opening of the East Sea (Otofuji et al., 1985, 1991). Under the assumption of a rigid block, this model indicates that rapid clockwise and counterclockwise rotations (about 40°) of SW and NE Japan relative to Eurasia occurred around 16–14 Ma (Otofuji et al., 1991). The rotation pole of SW Japan is located at 129°E, 34°N (west of Kyushu Island, Japan) with an angular velocity of about 20°/Ma.

Hayashida et al. (1991) showed that the timing of rotation of SW Japan between 16 and 14 Ma was a part of the events in the earlier formation of the East Sea from 19 to 14 Ma (Tamaki et al., 1989) and suggested a drift of SW Japan before 16 Ma that involved two consecutive stages: (1) simultaneous drift of SW Japan before 16 Ma mainly by translation in the Japan Basin, and (2) clockwise rotation of SW Japan between 16 and 14 Ma by extension in the southwestern East Sea including the Yamato Basin. This two-stage opening hypothesis is consistent with the results of seismological (Katao, 1988) and volcanologic studies (Shimazu et al., 1990).

Jolivet et al. (1991) noticed a dextral shear zone over a width of about 500 km from east to west, and suggested that the deformation proceeded along two N–S-trending boundary shear zones with rotations of blocks between them. They proposed a model of right-lateral pull-apart deformation with clockwise rotations of rigid blocks in the southern transtensional domain and counterclockwise rotations in the transpressional domain. The change of spreading direction of the Pacific Plate (43 Ma-elbow in the Hawaiian–Emperor chain; Gordon, 1991) might have formed a right-lateral stress field in the area during the Eocene. Paleomagnetic data, however, show that the opening of the East Sea occurred in the Early to Middle Miocene and seems to be related to the Eocene volcanic activity in this area. Alternatively, there is a possibility that the continental collision of India and Asia may feed this stress field (Tapponnier et al., 1982).

By the magnetic surveys in the East Sea, the dominant feature of the ENE–WSW magnetic anomaly lineation trend was revealed. However, the fit was not typically good because of a weak anomaly amplitude of 200–300 nT especially in the western part of the East Sea (Isezaki and Uyeda, 1973). Based on the magnetic anomaly lineations, Isezaki (1986) proposed that the opening started at 19 Ma and ceased at 15 Ma in the Japan Basin when the half spreading rate was assumed to be 3.4 cm/year. Kono (1986) found that a typical wavelength of 30 to 50 km in the eastern part of the East Sea represents a fluctuation of Miocene geomagnetic reversal polarity of 0.3–0.5 Ma scale when the spreading rate was assumed as about 10 cm/year. He pointed out that the difficulty of identifying the magnetic profile in the East Sea because of the possibilities of spreading from multiple ridges or frequent ridge jumps, a distorted skewness by later thermal events or block motion in the crust, and a deep or thin magnetic source layer. He proposed a one-sided spreading model for the East Sea opening during Anomalies 5B (15 Ma) to 5AA (13 Ma) with a ridge jump at 5AC (14 Ma).

6. Different timings of clockwise rotations in southeastern Korea and SW Japan

The deflection of declination in southeastern Korea has been attributed to the dextral movement of the Yangsan Fault correlating the opening of the East Sea (Kim and Kang, 1989; Kim et al., 1993; Kikawa et al., 1994; K.H. Kim et al., 1994). This is also comparable to the deflection of declination from SW and NE Japan (Hayashida et al., 1991; Jolivet et al., 1991).

Based on paleomagnetic and geochronologic data showing counterclockwise deflections of declination from Miocene rocks on the Goto and Tsushima Islands between the Korean Peninsula and Kyushu Island, Ishikawa and Tagami (1991) proposed that the rotation pole of SW Japan was located to the east of the Tsushima area. They concluded that counterclockwise rotations of these islands took place at about 15 Ma by sinistral movement of the Tsushima–

Goto Tectonic Line, constraining the western margin of the clockwise rotated block of SW Japan. The sinistral movement resulted in compressive deformation in the Goto and Tsushima areas. They proposed the possibility of the drifting of SW Japan prior to clockwise rotation at about 15 Ma.

The Pohang area comprising the Yangnam and Yeonil Basins is bounded by the Yangsan Fault in the northwest, the Ulsan Fault in the southwest, the Gampo Fault in the southeast, and East Korean Fault in the northeast (Figs. 1 and 8). Both basins are fragmented by many faulted blocks (Fig. 8). Early Miocene volcanic activities (Yangbuk Group) are typical in this area. The dextral movement of the Yangsan Fault is a well known topic in paleo-stress field studies (Hwang, 1994), and also its clockwise rotations with vertical axes. The current study shows that the mean declination (1.7ºE, Ltc) of the Yeonil Group is indistinguishable from the present axial dipole direction, while that from the Yangbuk Group (35.1º) and the Eocene Volcanic Rocks (38.0º) implies a considerable clockwise deflection. It is possible that the dextral movement of the Yangsan Fault was associated with the large clockwise block rotations of the Pohang area. Hence the time of occurrence of the dextral movement of the Yangsan Fault is indicative of the onset time of the Yeonil Basin formation, at about 17.3 Ma, which occurred prior to the clockwise rotation of SW Japan about 15 Ma.

7. Tectonic evolution of the Uleung Basin

The geologic structure of the Uleung Basin, one of three major basins in the East Sea (Fig. 1), has been little investigated (Chough and Barg, 1987). Korea–Russia collaborative seismic work (H.J. Kim et al., 1994) concluded that the crustal layer of the Uleung Basin (with the thickest sedimentary sequence in the East Sea and typical P1, P2, and P3 wave velocities of Oceanic layers 2 (5.0–6.0 km/s) and 3 (6.4–7.1 km/s)) is more similar to those of the Yamato Basin (5.1–5.4 km/s and 6.7–7.2 km/s, respectably) than to those of the Japan Basin (4.5–5.6 km/s and 6.6–6.7 km/s).

Shimazu et al. (1990), on the basis of geochemistry and stratigraphy, proposed a single volcanic belt across Korea and Japan (called the Sado–Pohang Belt) situated on the south side of the Yamato and Uleung Basins. Tectonic evolution and volcanic activity of the Sado–Pohang Belt were related with those of the Yamato and Uleung Basins.

We agree that the spreading was simultaneously developed with crustal thinning of the East Sea; they both originated from multiple spreading centers as suggested by Kono (1986). The SSE spreading direction is estimated from the ENE–WSW magnetic lineation in the East Sea. The spreading rates of the East Sea might be dependent on the type of spreading ridge (one- or two-sided) at that time (Kono, 1986). In both cases, the full spreading rate of 15 cm/year during 2 m.y. is a little high when we consider the drift distance of about 300 km of oceanic and crustal thinning portions in the East Sea (Hayashida et al., 1991).

8. Tectonic correlation of the dextral movement of the Yangsan Fault with the spreading of the East Sea

Tectonic correlation of the dextral movement of the Yangsan Fault with the spreading of the Uleung Basin is still poorly understood. The spreading of the Uleung Basin along the East Sea Fault can result in a transtensional stress field in the southeastern part of the Korean Peninsula (Fig. 8). This stress field could move the Yangsan Fault dextrally. The dextral movement of the Yangsan Fault may generate an extension zone in the Pohang area, which develops normal faults (Fig. 8). The formation of the Yeonil Basin can be explained by this process. In conjunction with the subsidence of the basin, the sedimentation environment rapidly changed from terrestrial to oceanic conditions, which has been supported by microfossil studies (e.g., Yoon et al., 1991; Yun et al., 1997).

The current study suggests that rotations of the Pohang area, formed by dextral movement of the Yangsan Fault, occurred obviously prior to the rotations of SW and NE Japan (see above). Combined with Japanese results (two-stage opening in the East Sea, Hayashida et al., 1991; Ishikawa and Tagami, 1991), the dextral movement of the Yangsan Fault was activated during a short-term interval in the event at around 17.3 Ma, which developed by

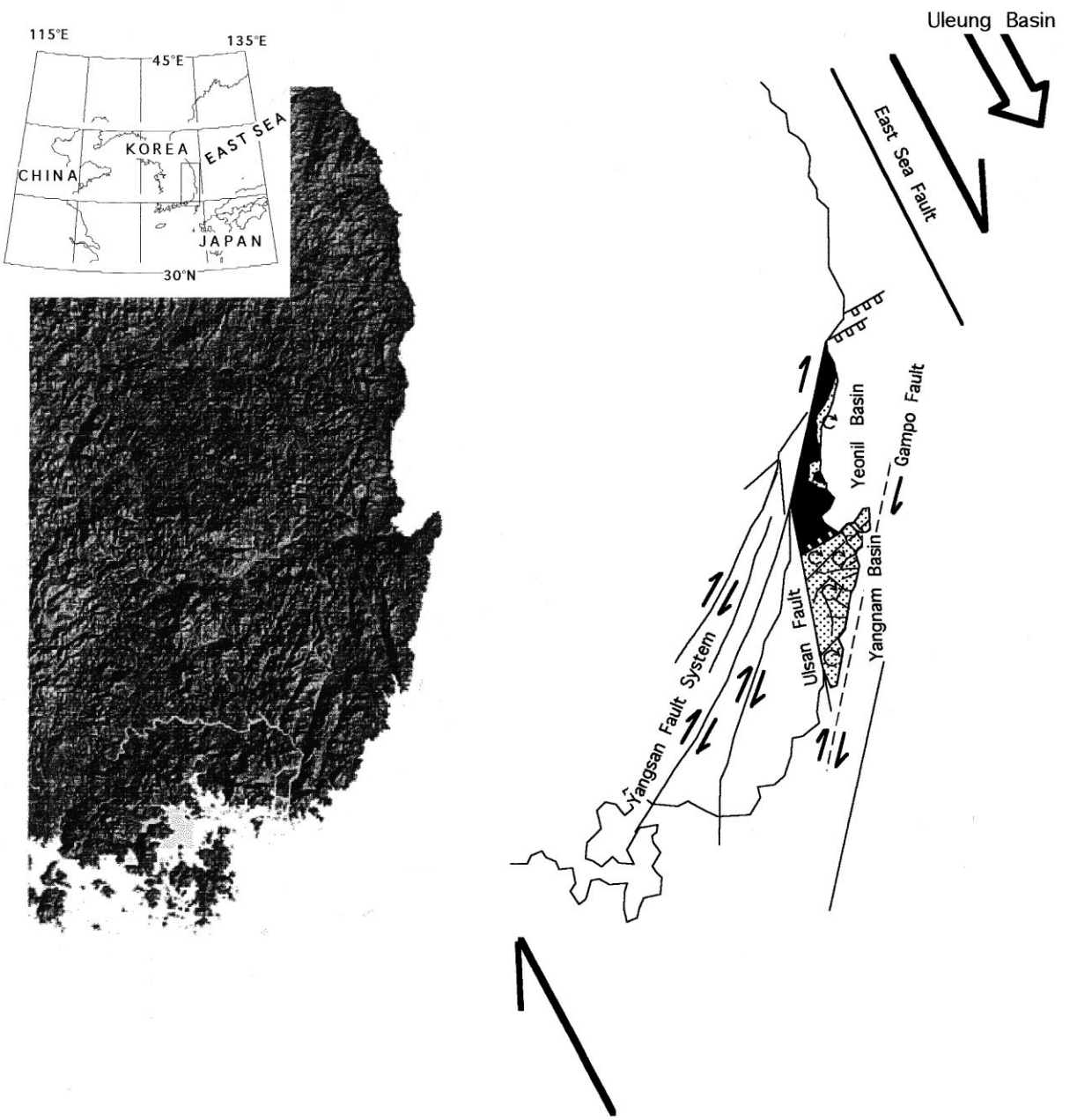


Fig. 8. (a) Topographic map, showing lineations of the study area. (b) The dextral Yangsan Fault formed by spreading of the Uleung Basin generates extensional faults (Yeonil Basin): gray, clockwise rotated blocks; black, newly created basin (non-rotated part). For further explanation, see text.

spreading of the Uleung Basin in western part of the East Sea in a SSE direction (Figs. 8 and 9). From a structural study of the Pohang area, Han et al. (1986) concluded that the Yeonil Basin was pulled apart by the dextral movement of the Yangsan Fault. We agree with their interpretation, and suggest that the movement was activated at about 17.3 Ma by the SSE directional spreading of the Uleung Basin.

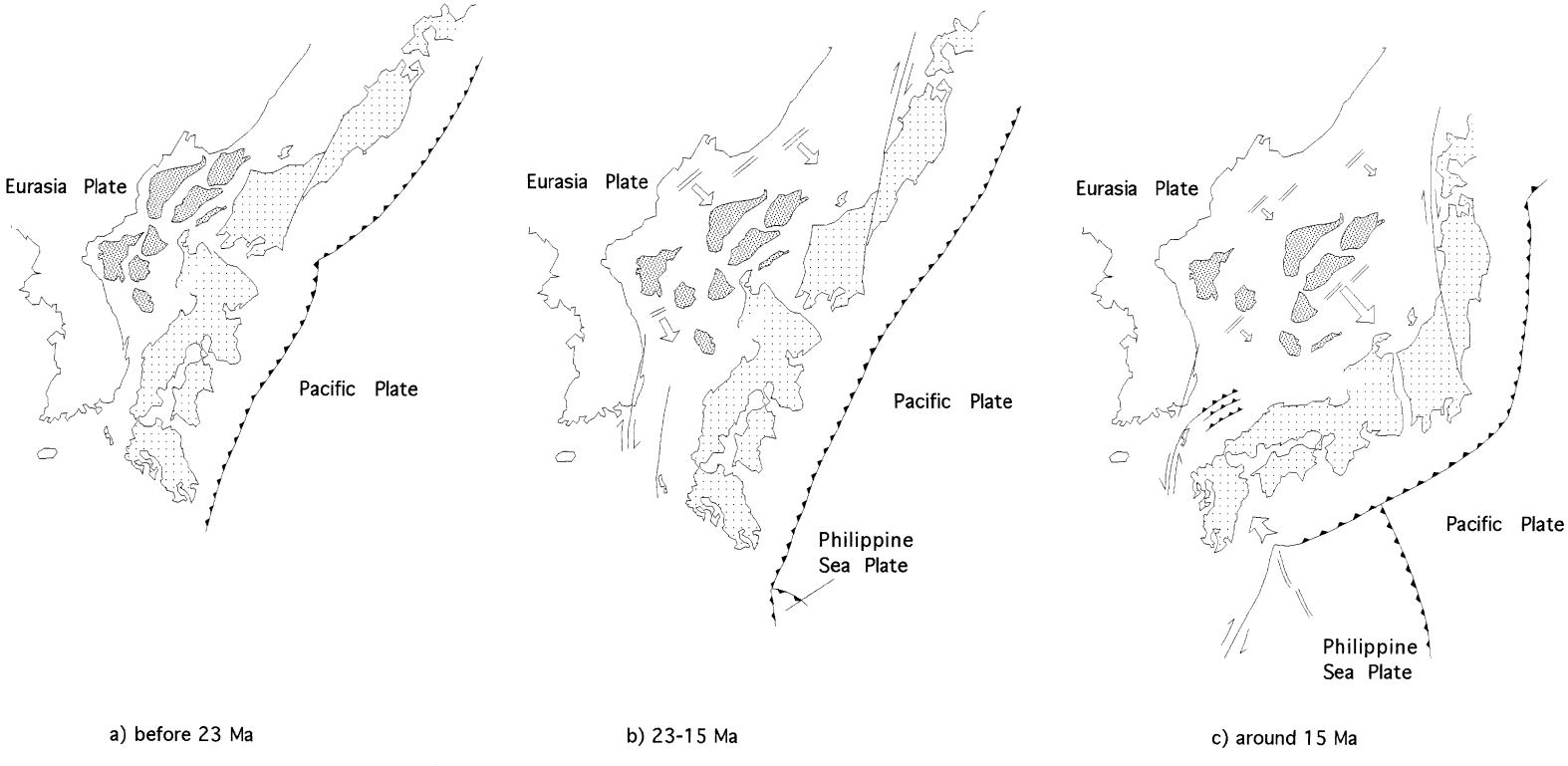


Fig. 9. The opening mode in the East Sea: (a) pre-opening stage, (b) back-arc spreading with crustal thinning and extensive volcanic eruption, (c) rotation of the Japanese Islands by pivoting of the Philippine Sea Plate. See Fig. 1 for tectonic lines and microblocks in the Sea and text for further explanation.

The spreading event around 17.3 Ma obtained for the Pohang area does not correspond with the entire spreading history of the East Sea. This may only indicate one of the spreading events in the East Sea. Isotopic data of green tuffs of around 23 Ma (Nohda and Wasserburg, 1986, Jin et al., 1988; Kimura et al., 1993) might be accepted as the initial stage of the East Sea opening (Fig. 9a). The southward drifts of the SW and NE Japan blocks were activated by the spreading and crustal thinning of the Japan, Yamato, and Uleung Basins between 23 and 15 Ma (Fig. 9b) before the rotations of the SW and NE Japanese blocks at around 15 Ma (Fig. 9c). More detailed geologic and geophysical information is still necessary for a realistic spreading model.

9. A probable mechanism of two stages of the opening

Paleomagnetic data indicating two stages of the opening in the East Sea give an additional clue for driving mechanism. It is noteworthy that the East Sea opened during two consecutive stages in Early Miocene time. The tectonic conditions of three plates (Eurasia, Pacific, and Philippine Sea plates) may be responsible for providing detailed explanations.

Haston and Fuller (1991) obtained valuable paleomagnetic data from around the Philippine Sea Plate, and suggested that the plate rotated clockwise up to 80º, moving about 20º northward since the Eocene. Hyodo and Niitsuma (1986) and Takahashi and Watanabe (1993) found an eastward declination deflection (during ca. 15–12 Ma) of the Kanto Syntaxis located in Central Japan indicating a lateral bending by collision between the Japan Island and the Izu Bonin Arc (in the Philippine Sea Plate). The plutonic and volcanic rocks are typical for the southern part of SW Japan (southern Kyushu–southern Shikoku–Shizuoka area), and are well known as a Miocene igneous activity around 17–10 Ma (Kimura et al., 1993). This reveals that the Philippine Sea Plate may have been already located near southern part of the SW Japan around 15 Ma. Recently, Hall et al., 1995a,b) and Hall (1996) obtained two Euler poles (10ºN/150ºE for 40–50 Ma and 15ºN/160ºE for 5–25 Ma) from paleomagnetic data for the Philippine Sea Plate, and provided reconstruction maps.

It is notable that the northeastward migration together with the clockwise rotation of the Philippine Sea Plate may play an important role as a pivot point in the area. A subduction zone of the northern extension for the Izu Bonin Arc is the most plausible pivot point. At about 15 Ma (fig. 4 in Hall, 1996), the paleo-pivot position was situated at about 32ºN/132ºE, which is near the Kyushu area in SW Japan. We believe that a pivot formed by collision between the Philippine Sea and Eurasia Plates was situated at the northern end of a ridge of the Philippine Sea Plate (approx. in the south of the Kyushu region) under the subduction zone (Fig. 9c), and that SW Japan was constrained to rotate clockwise for the Euler pole at the paleoposition of southern Kyushu.

The spreading rates became different from site to site in the East Sea. The spreading rate of the Yamato Basin was likely faster than that of the Japan or Uleung Basins, and the clockwise rotation of SW Japan and the counterclockwise rotation of NE Japan could have occurred around 15 Ma because of the collision with the Philippine Sea Plate. The sinistral Tsushima–Goto tectonic line is the most probable tectonic boundary in the west of SW Japan (Ishikawa et al., 1989). This idea is strongly supported by the considerable counterclockwise block rotations around 15 Ma based on paleomagnetic data from the Tsushima and Goto Islands (Ishikawa and Tagami, 1991) and the existence of compressive deformation and thrust belts, 100 km wide (Fig. 9c), about northwestern SW Japan.

10. Conclusions

Dual polarity paleomagnetic data obtained from Tertiary rocks in the southeastern Korean Peninsula indicate a close relationship with the opening of the East Sea.

A large clockwise block rotation by dextral movement of the Yangsan Fault on the Korean Peninsula occurred at 17.3 Ma, and is interpreted as revealing a parallel spreading event before a fan-shaped rotational event of the Japanese Islands (at around 15 Ma) during the opening of the East Sea.

Two stages of spreading (parallel at 23–15 Ma and pivotal at around 15 Ma) with crustal thinning are proposed in the East Sea area by combining data

from Korea and Japan. The pivotal stage found in Japan may be related to collision with the Philippine Sea Plate.

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